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Predicting Match Outcomes from Previous Performance: A Study of Momentum and Recovery in Soccer

Introduction

Sports are often discussed in terms of momentum, confidence, and recovery. After a team wins by a large margin, fans and analysts often ask whether that team will carry the momentum into the next game or become overconfident. After a major loss, the opposite question appears: will the team respond strongly, or will the loss hurt confidence and lead to another poor performance? These ideas are common in sports discussion, but they are difficult to measure because match outcomes are affected by many different factors.

The purpose of this project is to study whether previous team performance can reveal measurable patterns related to momentum and bounce-back behavior in soccer. Instead of only asking whether a team won or lost its previous match, this project looks at more detailed performance statistics from recent games. These include goal difference, shot difference, shot-on-target difference, corner difference, foul difference, and a generated performance score. The main goal is not to perfectly predict every soccer match, since soccer is too complex for that. Instead, the goal is to determine whether recent performance gives useful information about a team's next outcome.

This project focuses on Serie A, the highest professional soccer league in Italy. Serie A provides a strong dataset because it includes many seasons of match results and detailed game statistics. Each match includes information such as the home team, away team, final score, shots, shots on target, corners, fouls, and match result. These statistics make it possible to convert each match into team-level observations and study how each team's recent performance connects to its future results.

The main classification problem in this project is a three-class supervised learning problem: predicting whether a team's next match result will be a win, loss, or draw. It avoids a simpler win/not-win model and focuses only on the three direct match outcomes. Several models were tested, including Logistic Regression, Support Vector Machine, and Random Forest. The models were evaluated using validation accuracy, test accuracy, balanced accuracy, a confusion matrix, and repeated partitioning.

The larger goal of the project is to answer several sports-related questions using data: Does momentum exist in soccer? Do teams perform better after big wins? Do teams bounce back after losses? Are big losses followed by weaker performances? Which previous-performance features are most useful for predicting future outcomes? While the model does not perfectly predict games, the results show meaningful patterns in how teams perform after big wins and big losses. It is important to remember that the goal is not to predict win or loss, I'd be very rich if so. It is to find proven mathematical evidence to prove these patterns, instead of the gut feeling opinions that flow through the sports world.

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Description of Dataset(s)

The dataset used in this project was made up of Serie A match data from multiple CSV files. In the final multi-season run from the 2018/2019 season to the most recent 2025/2026 season, the program read 8 files, containing 3,000 total matches. After processing the matches into team-level observations, the dataset contained 5,904 team-match rows used for modeling. This transformation was important because each soccer match contains two teams, and each team needs its own row so that the model can study that team's previous performance and next result.

Each original row in the dataset represents one match. The important columns include the home team, away team, full-time home goals, full-time away goals, full-time result, home shots, away shots, home shots on target, away shots on target, home fouls, away fouls, home corners, and away corners. These columns were used to create team-level features for both the home and away team. Betting odds and other extra columns were irrelevant to this project so they were not included in any calculations and were completely ignored.

The target variable was the match result from the perspective of each team. For every team-match row, the result was labeled as Win, Loss, or Draw. This created a supervised classification problem. The final class distribution was fairly balanced between wins and losses, but draws occurred less often. There were 2,171 wins, 2,153 losses, and 1,580 draws. In percentage terms, wins made up 36.8% of the dataset, losses made up 36.5%, and draws made up 26.8%. This means the dataset was not extremely imbalanced, but draws were still the smallest class. In a complete team-match dataset, the number of wins and losses should be equal because every winning team corresponds to one losing team in the same match. The small difference between wins and losses in this project occurred because the first few matches for each team were removed before modeling. This was necessary because the model used previous-match history features, and early-season matches did not have enough prior games available. Since this filtering was done at the team level, one side of a match could occasionally be removed while the other side remained. The difference was very small, so it did not meaningfully affect the overall class balance.

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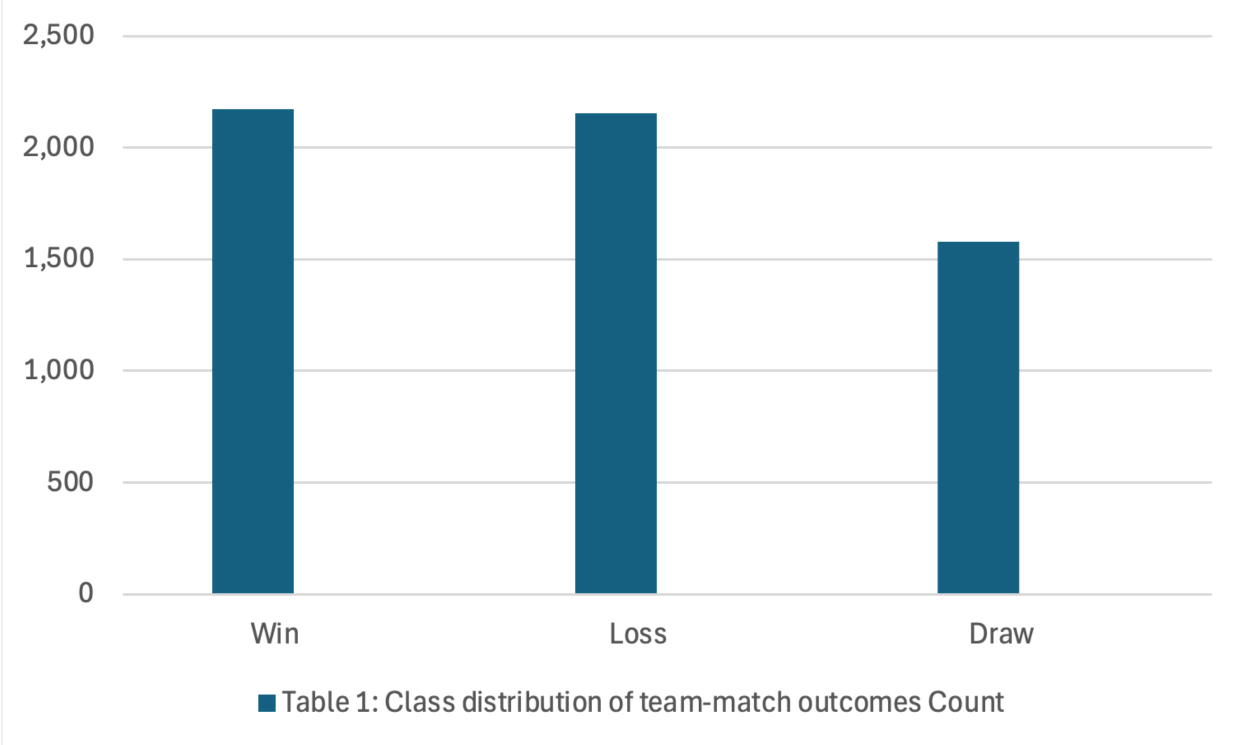

This class distribution matters because machine learning models often perform better on classes with more examples. Since draws were less common than wins or losses, the model had fewer examples to learn from. Also, draws are naturally difficult to predict because they can look statistically similar to both wins and losses. A team might dominate a match and still draw, or it might play poorly and still draw. This makes draws harder to separate using only previous-performance statistics.

The dataset is appropriate for the project because it includes many samples and real-world sports performance variables. It is not a toy dataset because match outcomes are noisy and affected by many hidden factors such as injuries, tactics, team rotation, travel, referee decisions, and luck. However, the available statistics still allow the project to study whether recent form and previous performance are connected to future match outcomes.

Description of Analysis Conducted

The analysis began by converting each match into two team-level rows: one row for the home team and one row for the away team. This allowed every team's performance to be analyzed separately. For example, if Juventus played Napoli, one row was created from Juventus's perspective and another row was created from Napoli's perspective. The target result for each row was labeled as Win, Loss, or Draw depending on that team's result.

The main engineered features were based on differences between a team's statistics and its opponent's statistics. These included GoalDiff, ShotDiff, ShotTargetDiff, CornerDiff, and FoulDiff. GoalDiff measured how many more or fewer goals the team scored compared to the opponent. ShotDiff measured the difference in total shots. ShotTargetDiff measured the difference in shots on target. CornerDiff measured the difference in corner kicks. FoulDiff measured the difference in fouls committed. These features were chosen because they are simple, understandable, and directly related to match performance.

A generated PerformanceScore was also created to summarize overall match performance. This score combined important indicators such as goal difference, shot difference, shots-on-target difference, and other match statistics. The purpose of the performance score was not to replace the final result, but to provide a more detailed measure of how strongly or weakly a team performed. For example, two teams might both win, but one team may win by dominating the match while another team may win narrowly. The performance score helps separate those cases.

The analysis focused on previous matches, not the current match. This was important because using the current match's goal difference or shot difference to predict the current result would leak the answer into the model. Instead, the project calculated rolling features from the team's previous matches. The main rolling windows were the previous 3 matches and the previous 5 matches. These windows were used to create features such as previous average goal difference, previous average shot difference, previous average performance score, previous win count, previous loss count, previous big win count, and previous big loss count.

The project also separated big wins from regular wins and big losses from regular losses. A big win represented a stronger result than a normal win, while a big loss represented a more damaging result than a normal loss. This was important because the project proposal focused on whether teams react differently after major results. A regular win may not affect momentum as strongly as a dominant win, and a regular loss may not affect future performance as much as a heavy loss.

After feature engineering, the dataset was split into training, validation, and test sets. The training set was used to fit the models. The validation set was used to compare models and tune parameters. The test set was held out until the end and used to evaluate the final selected model. This process follows the required project structure because the test set should represent unseen data that the model did not use during training or parameter selection.

Several classifiers were tested. The models included Logistic Regression with different normalization methods, Support Vector Machine, and Random Forest. Logistic Regression is useful because it is simple and interpretable. SVM is useful because it can model more complex boundaries between classes. Random Forest is useful because it can capture nonlinear patterns and feature interactions. Different parameters were tested for each model, and the validation set was used to select the best one.

Normalization was also tested. Some models were trained with min-max scaling, while others used standard z-score scaling. This was especially important for Logistic Regression and SVM because those models are sensitive to the scale of the features.

Random Forest does not rely on normalization in the same way, but it was still compared against the normalized models to see which approach performed best.

Finally, the project evaluated both predictive performance and momentum patterns. Predictive performance was measured using accuracy, balanced accuracy, a confusion matrix, and a classification report. Momentum was evaluated using bounce-back tables that compared future win rates and performance scores after big wins, regular wins, big losses, and regular losses. This allowed the project to go beyond simple accuracy and connect the results back to the original sports questions.

Presentation of Results

In the final multi-season experiment, the best model was Random Forest with a maximum depth of 14 and minimum samples per leaf of 3. This model achieved a validation accuracy of 0.447 and a validation balanced accuracy of 0.422. Logistic Regression with min-max scaling was close behind, with a validation accuracy of 0.447, but Random Forest had the best overall validation result.

On the final test set, the Random Forest model achieved a test accuracy of 0.438 and a test balanced accuracy of 0.412. This means the model predicted the correct result about 43.8% of the time. Since the task has three possible classes, this is better than random guessing, but it also shows that soccer outcomes are difficult to predict using only previous-performance statistics.

The confusion matrix showed that the model predicted wins and losses better than draws. For actual wins, the model correctly predicted 179 wins, while misclassifying 114 as losses and 33 as draws. For actual losses, the model correctly predicted 174 losses, while misclassifying 98 as wins and 51 as draws. For actual draws, the model correctly predicted only 35 draws, while predicting 87 as wins and 115 as losses.

Confusion matrix

	Pred_Win	Pred_Loss	Pred_Draw
Actual_Win	179	114	33
Actual_Loss	98	174	51
Actual_Draw	87	115	35

Figure 2. Confusion matrix for the final Random Forest model on the test set.

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The classification report confirmed this pattern. The model had an F1-score of 0.52 for wins and 0.48 for losses. However, the F1-score for draws was only 0.20. This shows that the model learned more useful patterns for wins and losses than for draws. Draws were the weakest class because they were often confused with both wins and losses.

The repeated partition test showed that the results were stable. Across 20 repeated partitions, the mean accuracy was 0.43143, with an accuracy variance of 0.00021. The mean balanced accuracy was 0.40350, with a balanced accuracy variance of 0.00023. This means the model's performance did not change drastically depending on the specific train/test split.

The bounce-back and momentum results were one of the most important parts of this project. Teams with a big win in the previous 3 matches had a next-match win rate of 44.5%, while teams with a regular win in the previous 3 matches and no big win had a next-match win rate of 34.1%. This created a win-rate gap of 10.5 percentage points in favor of teams with a recent big win.

The 5-match window showed a similar result. Teams with a big win in the previous 5 matches had a next-match win rate of 43.2%, while teams with a regular win in the previous 5 matches and no big win had a next-match win rate of 31.2%. This created a win-rate gap of 12.0 percentage points. These results suggest that big wins are connected to stronger future outcomes than regular wins.

The loss results showed the opposite pattern. Teams with a big loss in the previous 3 matches had a next-match win rate of 30.5%, while teams with a regular loss in the previous 3 matches and no big loss had a next-match win rate of 37.8%. This means recent big losses were followed by a lower chance of winning the next match.

The 5-match loss window also showed a negative pattern. Teams with a big loss in the previous 5 matches had a next-match win rate of 31.3%, while teams with a regular loss in the previous 5 matches and no big loss had a next-match win rate of 40.4%. This created a win-rate gap of -9.2 percentage points for teams with recent big losses. These results do not strongly support a bounce-back effect after big losses. Instead, they suggest that big losses may be connected to continued poor form.

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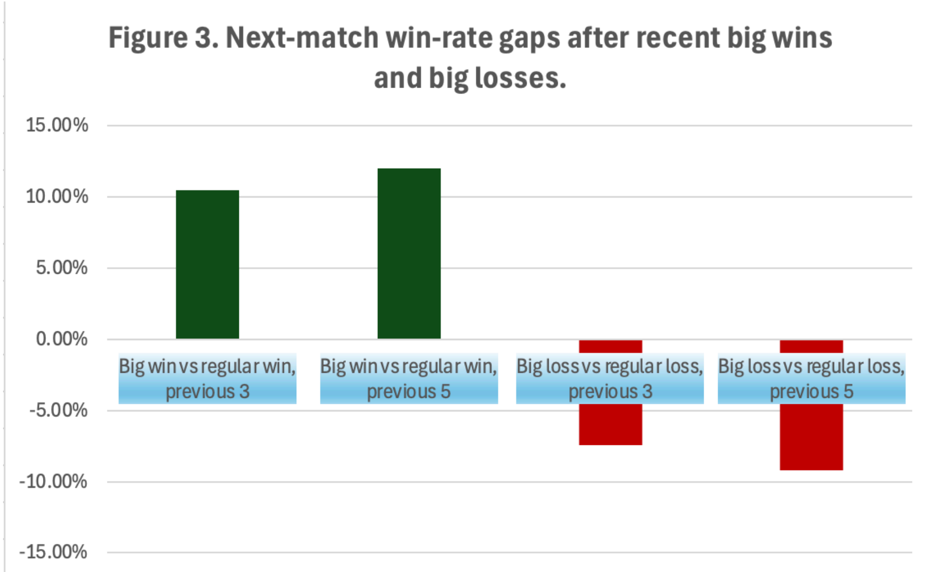

The feature importance results showed that the most important feature was the engineered Prev5PerformanceScoreAvg metric. This means that the average performance score over the previous five matches was the strongest predictor. Other important features included IsHome, Prev3LossCount, Prev3GoalDiffAvg, Prev3BigLossCount, Prev5FoulDiffAvg, Prev5LossCount, and Prev5ShotDiffAvg. These features support the idea that recent form over multiple matches matters more than only the immediately previous game. Also, interestingly being home was a large advantage, something that admittedly is a widely known phenomenon but it was interesting to prove it with calculations.

Analysis of Results

The results show that previous performance contains some useful information, but it is not enough to perfectly predict soccer outcomes. A test accuracy of 43.8% is moderate, but it is understandable for this problem. Soccer matches are low-scoring and can be affected by small events such as a red card, injury, penalty, missed chance, or tactical change. Many of these factors are not included in the dataset. Because of that, the model should not be judged only by accuracy. The more important part of this project is whether the model and engineered features reveal patterns related to momentum and recovery.

The model's strongest performance was on wins and losses. This makes sense because wins and losses often have clearer statistical patterns. A team that has been scoring more goals, creating more shots, and building stronger performance scores is more likely to look like a future winner. A team with repeated losses or poor goal differences is more likely to look like a future loser. The model was able to capture some of these patterns.

Draws were the most difficult outcome to predict. This was shown clearly in the confusion matrix and classification report. The model correctly predicted only 35 actual draws out of 237 draw examples in the test set. Most draws were predicted as either wins or losses. This is reasonable because draws can happen in many different ways. A draw can result from two evenly matched teams, a strong team underperforming, a weak team defending well, random finishing luck, or a team playing well but their shots keep hitting the post. Because of this, draws may not have a consistent pattern in previous-performance statistics.

Class imbalance also had some impact on the results. The dataset was not severely imbalanced, but draws were still less common than wins and losses. Since draws made up only 26.8% of the data, the model had fewer draw examples to learn from. This likely contributed to the poor draw F1-score. However, the issue is not only class imbalance. Draws are also conceptually harder to define because they often sit between wins and losses statistically.

The bounce-back and momentum results were more meaningful than the raw classification accuracy. The results showed that big wins were followed by better future performance than regular wins. In both the 3-match and 5-match windows, teams with recent big wins had noticeably higher next-match win rates and higher next-match performance scores. This suggests that big wins may reflect real team strength, confidence, or positive momentum.

The results for big losses were also important. Teams with recent big losses had lower next-match win rates than teams with regular losses. This suggests that big losses did not usually create a strong bounce-back effect in the data. Instead, big losses were more often connected with continued weak performance. This could mean that heavy losses are not just random bad games. They may reveal deeper issues such as poor form, defensive weakness, injuries, lack of confidence, or a difficult schedule.

The feature importance results also support the project's main idea. The most important feature was the average performance score over the previous five matches. This suggests that performance over a longer recent window matters more than just one match. Features such as previous loss count, previous goal difference, and previous big loss count were also important. This means the model was using form and momentum-related features, not just random noise.

The home-field feature was also important. This is expected because soccer teams often perform better at home. Home advantage can come from crowd support, less travel, familiarity with the stadium, and tactical comfort. Including the home/away context helped the model separate outcomes more effectively.

Overall, the results suggest that momentum exists in the data, but it should be interpreted carefully. The model does not prove that psychology alone causes better or worse results. A big win may be a sign of confidence, but it may also simply be a sign that the team is already strong. A big loss may hurt confidence, but it may also reveal that the team is weaker or facing tougher opponents. Therefore, the results support the idea that recent performance patterns matter, but they do not fully explain why those patterns occur.

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Conclusion

This project studied whether previous team performance can help predict future soccer outcomes and reveal patterns of momentum or bounce-back behavior. The project used multi-season Serie A data and converted match-level rows into team-level observations. The final dataset included 3,000 matches and 5,904 team-match rows. The target variable was a three-class result: Win, Loss, or Draw.

The analysis used engineered performance features such as GoalDiff, ShotDiff, ShotTargetDiff, CornerDiff, FoulDiff, and PerformanceScore. Rolling features from the previous 3 and 5 matches were created to measure recent form. The project also compared big wins to regular wins and big losses to regular losses. This allowed the analysis to focus on the original question of whether teams carry momentum or bounce back after major results.

Several machine learning models were tested, including Logistic Regression, SVM, and Random Forest. The best model was Random Forest with `max_depth = 14` and `min_samples_leaf = 3`. It achieved a final test accuracy of 43.8% and a balanced accuracy of 41.2%. The model predicted wins and losses better than draws, while draws were the most commonly confused class.

The most important conclusion is that recent performance does show meaningful patterns. Teams with recent big wins had higher next-match win rates than teams with only regular wins. Teams with recent big losses had lower next-match win rates than teams with only regular losses. This suggests that big wins are associated with positive future performance, while big losses are associated with weaker future performance. This means that momentum is important and a large predictor for teams. The bounce-back psychology phenomenon is largely disproven in a general sense but it does not mean that it is not real in specific examples but from these features and when predicting a large number of games, it is not a reliable predictor.

The project did not find strong evidence that teams automatically bounce back after big losses. Instead, the data showed that big losses were usually followed by worse outcomes compared to regular losses. This may mean that heavy losses reflect deeper problems in team form rather than just one bad game.

In conclusion, soccer outcomes are difficult to predict, and previous-performance statistics alone cannot explain everything. However, the results support the idea that momentum-like patterns exist in soccer data. Big wins, big losses, recent performance score, recent loss count, and recent goal difference all provided useful information. The project shows that even when prediction accuracy is limited, machine learning can still help reveal meaningful behavioral and performance patterns in sports.

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